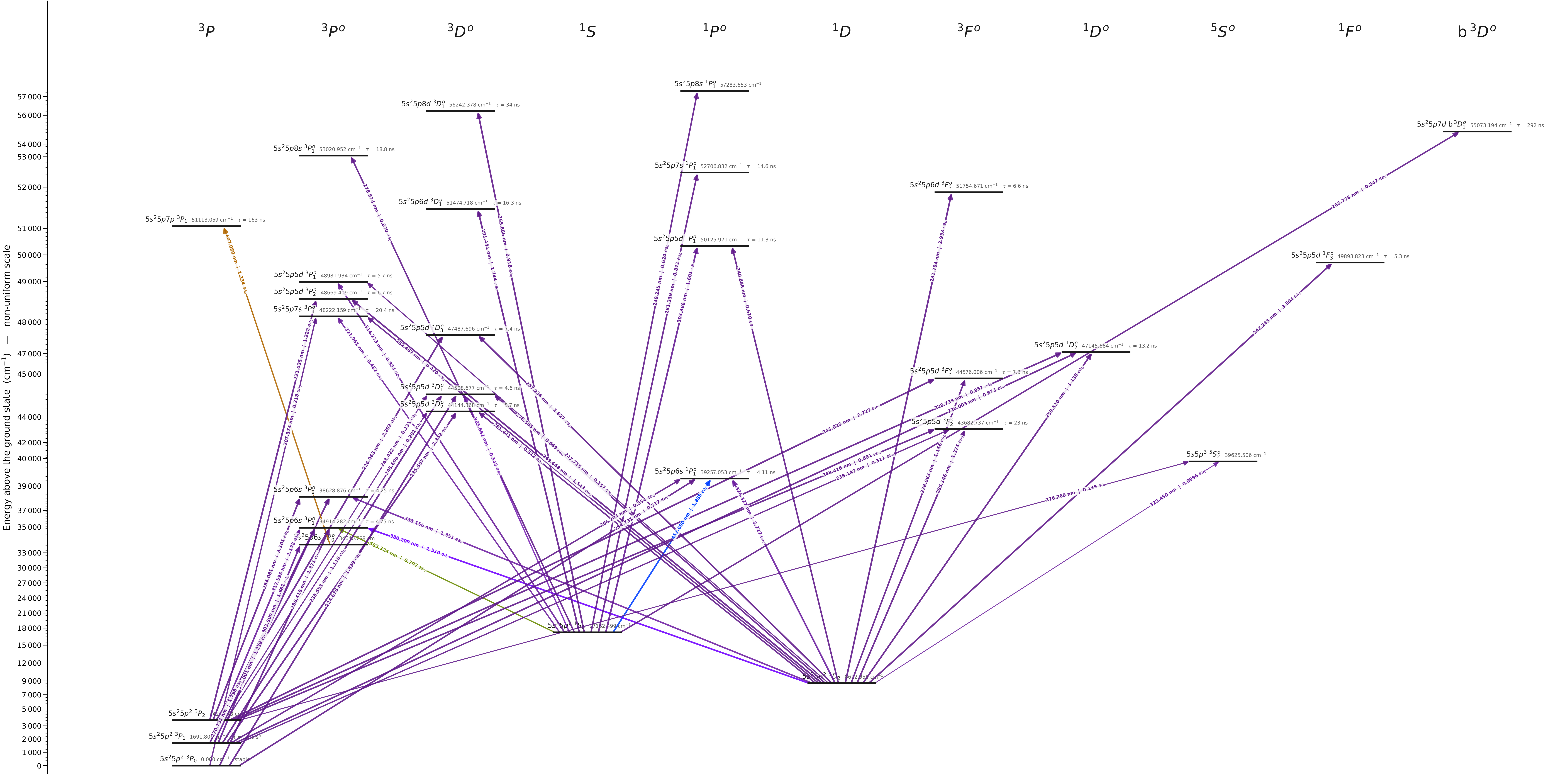
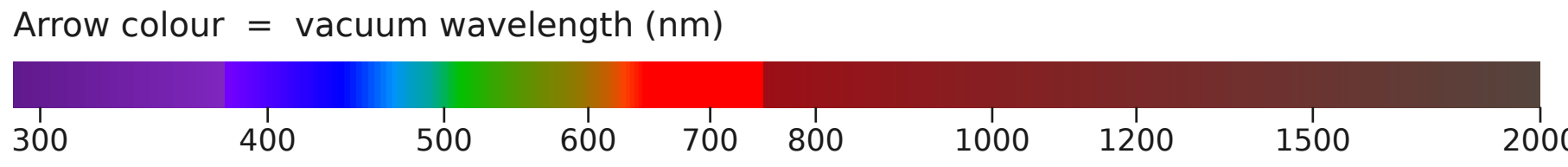


Tin

Sn (Sn I). Strongest classified transitions below 2 μm (NIST): 29 levels, 51 lines.

Sn⁺ ionisation limit 59232.69 cm⁻¹ = 7.34392 eV (168.826 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J||er||J'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s ² 5p ² ¹ S ₀ – 5s ² 5p7s ¹ P ₁ ^o	281.3388	281.2560	1065.5923	0.871	2.300e+07	10.9	33.6	alternative first step for collinear resonance ionization...
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.0812	283.9977	1055.3055	3.101	1.700e+08	37.45	72.2	first step of the resonance-ionization scheme used for co...
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₀ ^o	286.4157	286.3317	1046.7038	1.371	5.400e+07	33.51	25.7	ground-state resonance line used for high-resolution
5s ² 5p ² ³ P ₁ – 5s ² 5p6s ³ P ₁ ^o	303.4998	303.4115	987.7847	1.661	2.000e+08	–	–	proposed laser-cooling / magneto-optical-trap transition:...
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	317.5953	317.5035	943.9448	2.178	1.000e+08	33.51	47.5	strongest decay branch of the 5p6s 3P°1 level
5s ² 5p ² ¹ S ₀ – 5s ² 5p6s ¹ P ₁ ^o	452.6002	452.4733	662.3781	1.889	2.600e+07	38.72	10.7	visible line from the metastable 1S0 level used for colli...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

119Sn hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5s ² 5p ² ³ P ₁	578.296(4)	–	F=1/2: -578.30 F=3/2: +289.15
5s ² 5p ² ¹ D ₂	-1269.419(3)	–	F=3/2: +1904.13 F=5/2: -1269.42
5s ² 5p ² ¹ D ₂	-1328.745(4)	–	F=3/2: +1993.12 F=5/2: -1328.74
5s ² 5p6s ³ P ₁ ^o	-5011(3)	–	F=1/2: +5011.00 F=3/2: -2505.50
5s ² 5p6s ¹ P ₁ ^o	-259.1(10)	–	F=1/2: +259.10 F=3/2: -129.55

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5s ² 5p ² ¹ S ₀ – 5s ² 5p7s ¹ P ₁ ^o	281.339	116-112	491(8)
5s ² 5p ² ¹ S ₀ – 5s ² 5p7s ¹ P ₁ ^o	281.339	118-112	754(6)
5s ² 5p ² ¹ S ₀ – 5s ² 5p7s ¹ P ₁ ^o	281.339	122-112	1126(5)
5s ² 5p ² ¹ S ₀ – 5s ² 5p7s ¹ P ₁ ^o	281.339	124-112	1276(5)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	114-112	314(5)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	115-112	417.2(22)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	116-112	655(5)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	117-112	755(4)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	118-112	958(3)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	119-112	1048(3)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	120-112	1225.3(22)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	122-112	1460(3)
5s ² 5p ² ³ P ₂ – 5s ² 5p6s ³ P ₂ ^o	284.081	124-112	1678(3)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	114-112	319.5(16)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	115-112	412.7(15)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	116-112	640.0(12)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	117-112	746.8(12)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	118-112	945.2(12)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	119-112	1039.3(15)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	120-112	1214.4(11)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	122-112	1448.5(12)
5s ² 5p ² ³ P ₀ – 5s ² 5p6s ³ P ₁ ^o	286.416	124-112	1656.0(12)

Shift = ν(first isotope) – ν(second isotope).